

1: Factorial Calculator with Exception Handling

Problem Statement

Create a class `FactorialCalculator` with a method:

```
long factorial(int n)
```

This method computes the factorial of a number.

Exception Conditions

- If `n` is negative → throw exception with message:
"Number should not be negative"
- If `n` is greater than 20 → throw exception with message:
"Number too large to compute factorial safely"

Input Format

Each line contains a single integer `n`.

Output Format

- Print factorial result if valid.
- Otherwise print the exception message.

Sample Input

```
5
-3
21
```

Sample Output

```
120
java.lang.Exception: Number should not be negative
java.lang.Exception: Number too large to compute factorial safely
```

2: Bank Withdrawal System

Problem Statement

Create a class `BankAccount` with method:

```
double withdraw(double balance, double amount)
```

Exception Conditions

- If `amount <= 0` → throw exception:
"Withdrawal amount must be positive"
- If `amount > balance` → throw exception:
"Insufficient balance"

Input Format

Each line contains `balance` and `amount`.

Output Format

- Print remaining balance if successful.
- Otherwise print exception message.

Sample Input

```
1000 200
500 600
300 -50
```

Sample Output

```
800.0
java.lang.Exception: Insufficient balance
java.lang.Exception: Withdrawal amount must be positive
```

3: Student Grade Validator

Problem Statement

Create a class `GradeValidator` with method:

```
char calculateGrade(int marks)
```

Logic

- 90–100 → A
- 75–89 → B
- 50–74 → C
- Below 50 → F

Exception Conditions

- If `marks < 0` or `marks > 100` → throw exception:
`"Invalid marks entered"`

Input Format

Each line contains an integer `marks`.

Output Format

- Print grade if valid.
- Otherwise print exception message.

Sample Input

```
95
67
120
-5
```

Sample Output

```
A
C
java.lang.Exception: Invalid marks entered
java.lang.Exception: Invalid marks entered
```

4: String Division Parser

Problem Statement

Create a class `StringDivider` with method:

```
int divide(String a, String b)
```

This method converts strings to integers and performs division.

Exception Conditions

- If input is not a valid integer → throw exception:
`"Invalid number format"`
- If division by zero → throw exception:
`"Cannot divide by zero"`

Input Format

Each line contains two strings `a` and `b`.

Output Format

- Print result if valid.

- Otherwise print exception message.

Sample Input

```
10 2
8 0
a 5
```

Sample Output

```
5
java.lang.Exception: Cannot divide by zero
java.lang.Exception: Invalid number format
```